

Solution Requirements

Date	15 March 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The administrator must securely log in using a username and password to access the dashboard and prediction system.	High
2	Authorization levels	Only the administrator is allowed to upload datasets, train the model, and view prediction results. Unauthorized users cannot access the system.	High
3	External interfaces	The system interacts with the trained Machine Learning model, CSV dataset, and web interface developed using Flask.	High
4	Transactions processing	The system receives applicant information, preprocesses the data, runs the ML model, and returns an approval or rejection prediction.	High

5	Reporting	The dashboard displays prediction results, model accuracy, confusion matrix, classification report, and applicant details.	Medium
6	Business rules, etc.	The prediction is based on applicant attributes such as income, employment status, credit history, loan amount, and other eligibility criteria used by the trained model.	High
7	Compliance to laws or regulations	Applicant data must be handled securely, used only for prediction purposes, and stored according to applicable data privacy and security practices.	Medium
8	Other	The system should validate user input before prediction and display meaningful error messages for invalid or missing values.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system should process applicant data and display the prediction quickly.</i>	Prediction result displayed within 2 seconds .

2	Scalability	The system should support increasing numbers of prediction requests and larger datasets without significant performance degradation.	Supports 1000+ prediction requests per day .
3	Security & Data Privacy	Applicant information must be protected from unauthorized access using secure authentication and proper data handling practices.	Only authenticated users can access the dashboard and applicant data.
4	Reliability & Availability	The application should remain available and provide consistent predictions with minimal downtime.	99% system availability during normal operation.
5	Usability & Accessibility	The web interface should be simple, responsive, and easy to use on desktops and mobile devices.	Users can complete a prediction in less than 3 minutes without training.
6	<i>Other</i>	The system should allow easy updates to the machine learning model, dataset, and application code without affecting existing functionality.	New model or dataset can be integrated with minimal code changes .